

HW

Ex 1.5:-

1. $f(x) = x, \quad g(x) = \sqrt{x-1}$

i. $(f+g)(x) = f(x) + g(x)$

$$= x + \sqrt{x-1}$$

— (1)

↓

Domain: $[1, \infty)$

Domain: $[-\infty, \infty]$

both intersection

Domain: $[1, \infty)$

Put '1' in eq-1 for range.

$$= 1 + \sqrt{1-1}$$

$$= 1 + \sqrt{0} = 1$$

$\Rightarrow$ Range: $[1, \infty)$

ii. $(f \cdot g)(x) = f(x) \cdot g(x)$

$$= x \sqrt{x-1} \quad \text{— (2)}$$

Domain: $[1, \infty)$

Put '1' in eq-2 for range.

$$= 1 \sqrt{1-1}$$

$$= 1 \sqrt{0}$$

$$= 0$$

Range: $[0, \infty)$

2. $f(x) = \sqrt{x+1}$, $g(x) = \sqrt{x-1}$.

i. $(f+g)(x) = f(x) + g(x)$

$= \sqrt{x+1} + \sqrt{x-1}$ - (3)

Domain: $[-1, \infty)$

Range: $[\frac{1}{2}, \infty)$

Domain: $[1, \infty)$

Put '1' in eq-3 for range:

$= \sqrt{1+1} + \sqrt{1-1}$

$= \sqrt{2} + \sqrt{0}$

$= \sqrt{2}$

Range: $[\sqrt{2}, \infty)$

ii. $(f \cdot g)(x) = f(x) \cdot g(x)$

$= \sqrt{x+1} \cdot \sqrt{x-1}$

$= \sqrt{x^2-1}$

Domain: $[1, \infty)$

Put '1' in eq-3 for range.

$\sqrt{1^2-1} = \sqrt{1-1} = \sqrt{0} = 0$

Range: $[0, \infty)$

$f(x) = 2$, $g(x) = x^2+1$

Domain: $(-\infty, \infty)$

Domain: $(-\infty, \infty)$

i. $(f/g)(x) = f(x)/g(x)$ where $g(x) \neq 0$

~~$= 2/(x^2+1)$~~

x^2+1

$\Rightarrow x^2+1 \neq 0$
 $\Rightarrow x^2 \neq -1$
 $\Rightarrow x \neq \sqrt{-1}$

Domain: $\mathbb{R}$

Range: $[0, 2]$

ii. $(f/g)(x) = g(x)/f(x)$, where $f(x) \neq 0$

$= x^2+1$

≥ 2

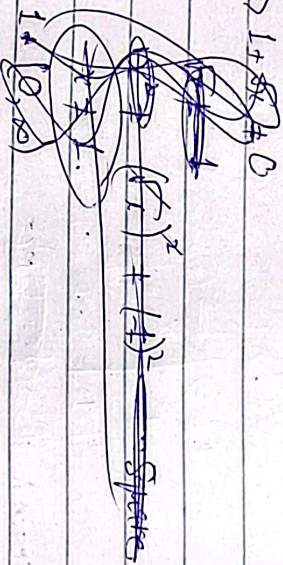
Domain: $(-\infty, \infty)$

Range: $[\frac{1}{2}, \infty)$

4. $f(x) = 1, g(x) = 1 + \sqrt{x}$

i. $\left(\frac{f}{g}\right)(x) = \frac{1}{1 + \sqrt{x}}$

$\Rightarrow 1 + \sqrt{x} \neq 0$



Domain: $[0, \infty)$

Range: $(0, 1]$

ii. $\left(\frac{g}{f}\right)(x) = 1 + \sqrt{x}$

Domain: $[0, \infty)$

Range: $[1, \infty)$

✓ 5. $f(x) = x + 5, g(x) = x^2 - 3$

a. $f(g(x)) =$

$g(x) = x^2 - 3$

$\Rightarrow g(10) = 10^2 - 3, \text{ so } \boxed{g(10) = 97}$

$f(g(10)) = f(97)$

$= -3 + 5$

$\boxed{f(g(10)) = 2}$

b. $g(f(x)) = g(x + 5)$

$= (x + 5)^2 - 3$

$\therefore f(10) = 0 + 5$

$= 5$

$\boxed{g(f(10)) = 22}$

c. $f(g(x)) = f(x^2 - 3)$

$= x^2 - 3 + 5$

$= x^2 + 2$

d. $g(f(x)) = g(x + 5)$

$= (x + 5)^2 - 3 = x^2 + 25 + 10x - 3$

$= x^2 + 10x + 22$

e. $f(f(-5)) = f(0)$

$= 0 + 5$

$\therefore f(-5) = -5 + 5 = 0$

$= 5$

f. $g(g(2)) = g(1)$

$= 1^2 - 3$

$\therefore g(2) = 2^2 - 3 = 1$

$= -2$

$$\checkmark g. f(f(x)) = f(x+5)$$

$$= x+5+5$$

$$= x+10$$

$$\checkmark h. g(g(x)) = g(x^2-3)$$

$$= (x^2-3)^2-3$$

$$= x^4+9-6x^2-3$$

$$6. f(x) = x-1, g(x) = \frac{1}{x+1}$$

$$= x^4-6x^2+6$$

$$a. f(g(\frac{1}{2})) = f(\frac{2}{3})$$

$$g(\frac{1}{2}) = \frac{1}{\frac{1}{2}+1}$$

$$= \frac{2}{3}-1 = \frac{2-3}{3}$$

$$= \frac{1}{3}$$

$$= -1/3$$

$$= \frac{2}{3}$$

$$b. g(f(\frac{1}{2})) = g(-\frac{1}{2})$$

$$f(\frac{1}{2}) = \frac{1}{2}-1 = -\frac{1}{2}$$

$$= \frac{1}{-\frac{1}{2}+1} = \frac{1}{1/2}$$

$$= 2$$

$$c. f(g(x)) = f\left(\frac{1}{x+1}\right)$$

$$= \frac{1}{\frac{1}{x+1}} - 1 = \frac{x+1}{1} - 1 = x$$

$$= \frac{-x}{x+1}$$

$$= \frac{-x}{x+1}$$

$$d. g(f(x)) = g(x-1)$$

$$= \frac{1}{x-1+1}$$

$$= \frac{1}{x}$$

$$e. f(f(x)) = f(1)$$

$$f(1) = 1-1 = 0$$

$$f. g(g(x)) = g(\frac{1}{x})$$

$$g(\frac{1}{x}) = \frac{1}{\frac{1}{x}+1} = \frac{x}{1+x}$$

$$= \frac{1}{\frac{1}{3}+1} = \frac{1}{\frac{4}{3}} = \frac{3}{4}$$

$$g. f(f(x)) = f(x-1)$$

$$= x-1-1 = x-2$$

$$h. g(g(x)) = g\left(\frac{1}{x+1}\right)$$

$$= \frac{1}{\frac{1}{x+1} + 1} = \frac{1}{\frac{1+x+1}{x+1}}$$

$$= \frac{x+1}{x+2}$$

$$7. u(x) = 4x - 5 \quad v(x) = x^2 \quad f(x) = \frac{1}{x}$$

$$a. u(v(f(x))) =$$

$$= v(f(x)) = v\left(\frac{1}{x}\right) = \frac{1}{x^2}$$

$$u\left(\frac{1}{x^2}\right) = 4\left(\frac{1}{x^2}\right) - 5 = \frac{4 - 5x^2}{x^2}$$

$$b. u(f(v(x))) =$$

$$= f(v(x)) = f(x^2) = \frac{1}{x^2}$$

$$u\left(\frac{1}{x^2}\right) = \frac{4 - 5x^2}{x^2}$$

$$c. v(u(f(x))) =$$

$$u(f(x)) = u\left(\frac{1}{x}\right)$$

$$= 4\left(\frac{1}{x}\right) - 5 = \frac{4 - 5x}{x}$$

$$v\left(\frac{4 - 5x}{x}\right) = \left(\frac{4 - 5x}{x}\right)^2 = \frac{16 - 40x + 25x^2}{x^2}$$

$$d. v(f(u(x))) =$$

$$= f(u(x)) = f(4x - 5)$$

$$= \frac{1}{4x - 5}$$

$$v\left(\frac{1}{4x - 5}\right) = \left(\frac{1}{4x - 5}\right)^2$$

$$e. f(u(v(x))) =$$

$$= u(v(x)) = u(x^2) = 4x^2 - 5$$

$$f(4x^2 - 5) = \frac{1}{4x^2 - 5}$$

$$4x^2 - 5$$

$$f. f(v(u(x))) =$$

$$\begin{aligned} \therefore v(u(x)) &= v(4x-5) \\ &= (4x-5)^2 \end{aligned}$$

$$8. f(x) = \sqrt{x}, g(x) = \frac{x}{4}, h(x) = 4x-8$$

$$a. h(g(f(x))) =$$

$$\therefore g(f(x)) = g(\sqrt{x}) = \frac{\sqrt{x}}{4}$$

$$h\left(\frac{\sqrt{x}}{4}\right) = 4\left(\frac{\sqrt{x}}{4}\right) - 8 = \sqrt{x} - 8$$

$$b. h(f(g(x))) =$$

$$\therefore f(g(x)) = f\left(\frac{x}{4}\right) = \sqrt{\frac{x}{4}}$$

$$h\left(\sqrt{\frac{x}{4}}\right) = 4\left(\sqrt{\frac{x}{4}}\right) - 8 = \sqrt{x} - 8$$

$$c. g(h(f(x))) =$$

$$\therefore h(f(x)) = h(\sqrt{x}) = 4(\sqrt{x}) - 8$$

$$g(4\sqrt{x}-8) = \frac{4\sqrt{x}-8}{4} = \sqrt{x}-2$$

$$1. g(f(h(x))) =$$

$$\begin{aligned} \therefore f(h(x)) &= f(4x-8) = \sqrt{4x-8} \\ g(\sqrt{4x-8}) &= \frac{\sqrt{4x-8}}{4} = \frac{\sqrt{4(x-2)}}{4} \end{aligned}$$

$$= \frac{1}{2}\sqrt{x-2}$$

$$c. f(g(h(x))) =$$

$$\begin{aligned} \therefore g(h(x)) &= g(4x-8) \\ &= \frac{4x-8}{4} = x-2 \end{aligned}$$

$$f(x-2) = \sqrt{x-2}$$

$$f. f(h(g(x))) =$$

$$\begin{aligned} \therefore h(g(x)) &= h\left(\frac{x}{4}\right) \\ &= 4\left(\frac{x}{4}\right) - 8 = x-8 \end{aligned}$$

$$f(x-8) = \sqrt{x-8}$$

$$11. (a) f(x) = \sqrt{x}, g(x) = x-7, f(g(x)) = ?$$

$$\text{f(x-7)}$$

$$\begin{aligned} f(g(x)) &= f(x-7) \\ &= \sqrt{x-7} \end{aligned}$$

b. $f(x) = 3x$, $g(x) = x+2$, $fg(x) = ?$

$$fg(x) = f(x+2) = 3(x+2) \\ = 3x+6.$$

c. $f(x) = \sqrt{x-5}$, $fg(x) = \sqrt{x^2-5}$, $g(x) = ?$

As $f(x) = \sqrt{x-5}$

$$\Rightarrow f(g(x)) = \sqrt{g(x)-5}$$

↓

$$\sqrt{x-5} = \sqrt{g(x)-5}$$

$$\Rightarrow x-5 = g(x)-5$$

So, $\boxed{g(x) = x^2}$

d. $g(x) = x$, $f(x) = x$, $fg(x) = ?$

$$fg(x) = f\left(\frac{x}{x-1}\right)$$

$$= \frac{x}{\frac{x}{x-1} - 1}$$

$$= \frac{x}{x-1}$$

$$x - (x-1)$$

$$= \frac{x}{x-1} = \frac{x}{x-1}$$

$$x - x + 1 \quad 0+1 \quad 1.$$

$$= x$$

e. $g(x) = ?$, $f(x) = \frac{1+x}{x}$, $fg(x) = x$.

As $f(x) = \frac{1+x}{x}$

$$\Rightarrow fg(x) = \frac{1+x}{g(x)}$$

↓

$$x = \frac{1+x}{g(x)}$$

$$x-1 = 1$$

$$g(x)$$

$$\Rightarrow \boxed{g(x) = \frac{1}{x-1}}$$

f. $g(x) = \frac{1}{x}$, $f(x) = ?$, $f \cdot g(x) = x$.

As $f \cdot g(x) = f\left(\frac{1}{x}\right) = x$

$\Rightarrow f\left(\frac{1}{x}\right) = x$

So, $f(x) = \frac{1}{x}$.



12(a) $g(x) = \frac{1}{x-1}$, $f(x) = |x|$, $f \cdot g(x) = ?$

$f\left(\frac{1}{x-1}\right) = \left|\frac{1}{x-1}\right|$

(b) $g(x) = ?$, $f(x) = x-1$, $f \cdot g(x) = x$

As $f(x) = x-1$

$\Rightarrow f(g(x)) = g(x)-1$

$\frac{x}{x+1} = 1 - \frac{1}{g(x)}$

$\Rightarrow \frac{1}{g(x)} = 1 - \frac{x}{x+1}$

$= \frac{x+1-x}{x+1}$

$\frac{1}{g(x)} = \frac{1}{x+1}$

So, $g(x) = x+1$

(c) $g(x) = ?$, $f(x) = \sqrt{x}$, $f \cdot g(x) = |x|$

As $f(x) = \sqrt{x}$

$\Rightarrow f \cdot g(x) = \sqrt{g(x)}$

$|x| = \sqrt{g(x)}$

$(|x|)^2 = (\sqrt{g(x)})^2$

$\Rightarrow \boxed{g(x) = x^2}$

(d) $g(x) = \sqrt{x}$, $f(x) = ?$, $f \cdot g(x) = |x|$

As $f \cdot g(x) = f(\sqrt{x}) = |x|$

$\Rightarrow \boxed{f(x) = |x^2|}$



✓ 13. $f(x) = \sqrt{x+1}$, $g(x) = \frac{1}{x}$

a) $f \circ g(x) = f\left(\frac{1}{x}\right) = \sqrt{\frac{1}{x} + 1}$

$f \circ g(x) = \sqrt{\frac{1+x}{x}} \rightarrow \text{Domain: } (0, \infty) \cup (-\infty, -1]$
 $\rightarrow \text{Range: } (1, \infty) \cup [0, 1]$

b) $g \circ f(x) = g(\sqrt{x+1}) = \frac{1}{\sqrt{x+1}}$
 $\rightarrow \text{Domain: } (-1, \infty)$
 $\rightarrow \text{Range: } (0, \infty)$

14. $f(x) = x^2$, $g(x) = 1 - \sqrt{x}$

a) $f \circ g(x) = f(1 - \sqrt{x}) = (1 - \sqrt{x})^2 = 1 + x - 2\sqrt{x}$

Domain: $[0, \infty)$

Range: $[1, \infty)$

b) $g \circ f(x) = g(x^2) = 1 - \sqrt{x^2} = 1 - |x|$

$g \circ f(x) = 1 - x$

Domain: $(-\infty, \infty)$

Range: $(-\infty, \infty)$

$y = -(x - (-7))^2 - 1$
 $y = -(x - 4)^2 - 1$

a) $y = x^2 + 3$

b) $y = x^2 - 5$

7. (a). Position 4. $y = (x - (-1))^2 + (-4)$

(b) Position 1. $y = (x - (-2))^2 + (2)$

(c) Position 2. $y = (x - (-2))^2 + (2)$

(d) " 3. $y = (x - (-3))^2 + (-2)$

18. (a) As per the given questions:

$y = (x - h)^2 + k$

(a) $V(1, 4) \rightarrow$ shift right, shift up 4.

$\Rightarrow y = -(x - 1)^2 + 4$

(b) $V(-2, 3) \rightarrow$ shift left 2, up 3.

$y = -(x - (-2))^2 + 3$

$\Rightarrow y = -(x + 2)^2 + 3$

(c) $V(-4, -1) \rightarrow$ shift left 4, up down 1

$y = -(x - (-4))^2 + (-1)$

$\Rightarrow y = -(x + 4)^2 - 1$

(d) $V(2,0) \rightarrow$ shift right 2.

$$y = -(x-2)^2$$

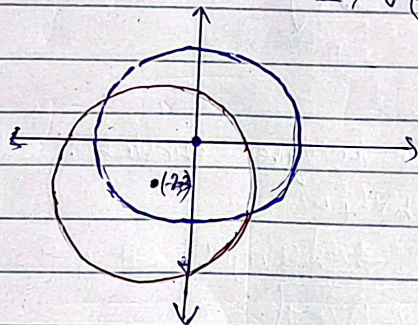


19. $x^2 + y^2 = 49$, down 3, left 2.

$$(x - (-2))^2 + (y - (-3))^2 = 49$$

$$\Rightarrow (x+2)^2 + (y+3)^2 = 7^2$$

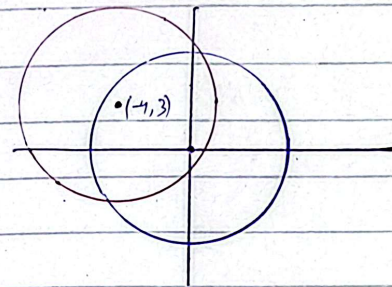
$$\Rightarrow V(-2, -3)$$



20. $x^2 + y^2 = 25$, up 3, left 4.

$$(x+4)^2 + (y-3)^2 = 25$$

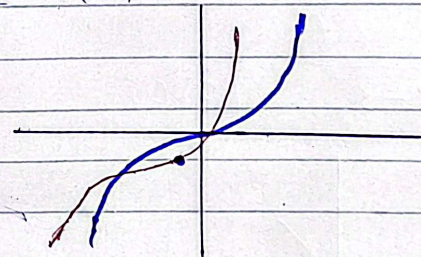
$$\Rightarrow V(-4, +3)$$



$y = x^3$, left 1, down 1.

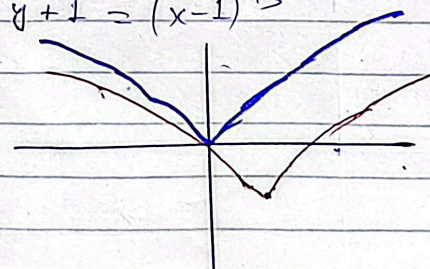
$$y+1 = (x+1)^3$$

$$\Rightarrow V(-1, -1)$$



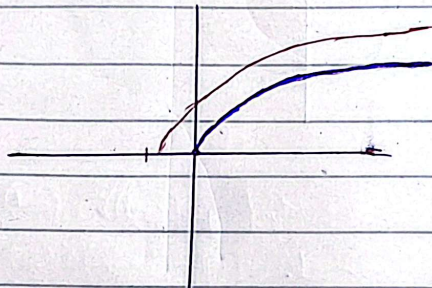
$y = x^{2/3}$, right 1, down 1.

$$\Rightarrow y+1 = (x-1)^{2/3} \Rightarrow V(1, -1)$$



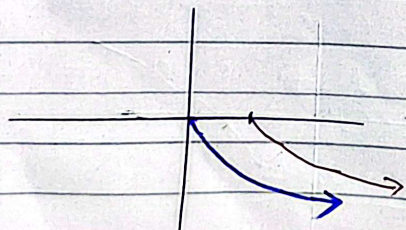
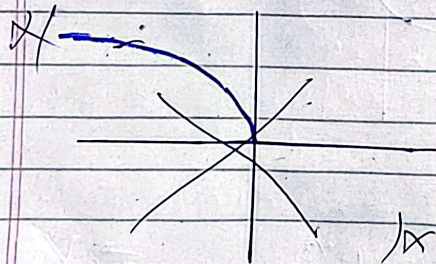
23. $y = \sqrt{x}$, Left 0.81

$y = \sqrt{x+0.81}$ $(-0.81, 0)$



24. $y = -\sqrt{x}$, Right 3.

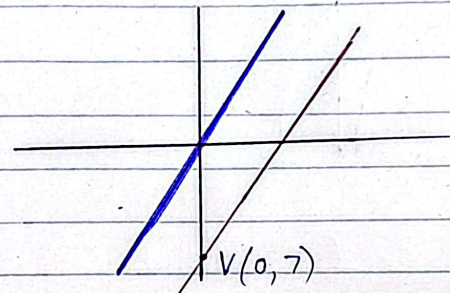
$y = -\sqrt{x-3}$ $(3, 0)$



$y = 2x - 7$, ^{up} Right = 7.

$y = -7 = 2x - 7 \Rightarrow V(0, 7)$

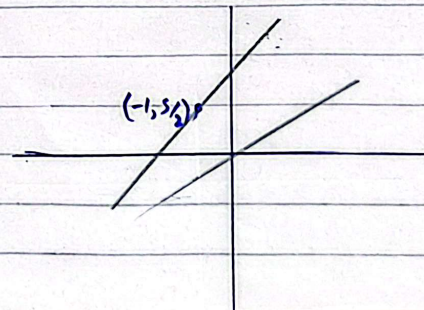
$\Rightarrow \boxed{y = 2x}$



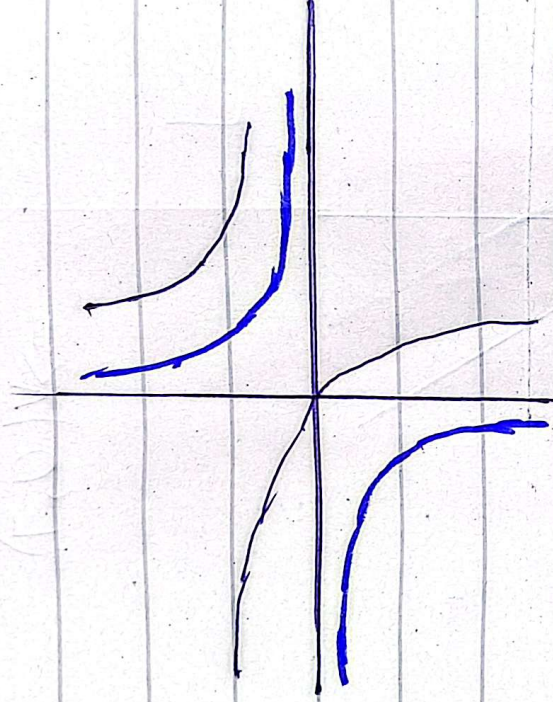
6. $y = \frac{1}{2}(x+1) + 5$, $\nearrow V(-1, 5/2)$
Down 5, right 1.

$y + 5 = \frac{1}{2}(x + 1 - 1) + 5$

$\Rightarrow y = \frac{1}{2}(x) \Rightarrow V(0, 0)$



27. $y = \frac{1}{x}$, Up 1, Right 1.



$$\Rightarrow y - 1 = \frac{1}{x - 1} \quad (1, 1)$$

28. $y = \frac{1}{x^2}$, Left 2, down 1.